Paper III: formalization and reproducibility

Juan Pablo Traverso Gianini
Independent researcher, Santiago, Chile
jtraverso@gmail.com
ORCID: 0009-0003-6068-4096

Paper III · Formal companion v1.6
19 October 2026

This document accompanies the mathematical article. It preserves the formal declarations, scope qualifications and computational evidence previously included in the article. The source freeze and its recorded verification are unchanged. No Lean/Lake build was run for this edition. Historical section and table numbers in the records below refer to the preceding article; they are retained as locators, not renumbered as new verification results. Original references [1], [2], etc. are listed at the end.

Frozen sources and historical evidence / Fuentes congeladas y evidencias históricas · Previous article / Artículo anterior

11.6 Formal-verification perimeter and current status

The formalization discussion is confined to this section and the release record in Section 13. The current formal snapshot separates the unconditional final assembly from its downstream interfaces. The public aggregate root PaperIII imports both PaperIII.Theorem_1_1_Final and PaperIII.PublicAPI. The final module discharges the two interfaces used by the paper and exports Theorem 1.1 and Corollary 1.2. PaperIII.CanonicalTrianglePacking makes PaperIII.nu3 and PaperIII.nu3Star the manuscript-facing quantities and proves their equality with the edge-hypergraph implementations used by AX1; it also proves PaperIII.tau3Star\=\PaperIII.nu3Star and identifies the cover-side AX1 interface with the packing-side statement in Section 2. PaperIII.PaperImprovements contains the paper-facing certificates; PaperIII.PublicAPI, PaperIII.OfPartition, and PaperIII.Obstructions expose downstream interfaces. The directed canonical gate reports the following axiom footprint for these theorem surfaces:

```
[propext, Classical.choice, Quot.sound]
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The final assembly uses explicit bridges to the Haxell–Rödl/Yuster and dense-decomposition formalizations. The cited asymptotic theorems remain in the manuscript for mathematical provenance. The recorded author-side public-root build completed 8,455 jobs with exit code 0; the subsequent query-root build completed 8,444 jobs with exit code 0. That author-side process was resumed after an application restart. Separately, the external audit performed an uninterrupted reproduction from an absent project build directory: the public root completed 8,455 jobs, the query roots completed 8,444 jobs, and all eight axiom-query files exited successfully. The immutable formal archive is sealed, and the internal and external Lean gates pass within the scope stated in Section 13.

Appendix E gives a curated selection of paper-level and reusable certificates. The complete implementation-level inventory remains in the formal source rather than being duplicated in the manuscript.

Table 2. Formalization surfaces. Statuses in this table are records of the sealed local freeze; they are not a substitute for independent reproduction.

Surface	Purpose	Current local status
PaperIII	aggregate/compatibility root	public-root build completed 8,455 jobs; exit 0; imports the final theorem and public API
PaperIII.MainNibble	theorem and corollary assembly through explicit AX1/AX2 interfaces	compiled in the recorded closure
PaperIII.Theorem_1_1_Final	unconditional discharge of AX1/AX2 and final theorem surfaces	compiled; final theorem footprint is foundational-only
PaperIII.CanonicalTrianglePacking	canonical nu3, nu3Star, tau3Star and AX1 statement bridges	compiled; exact target-side bridges recorded
PaperIII.CanonicalTrianglePackingGate	statement and axiom gate for the canonical interfaces	exit 0; foundational-only footprints recorded
PaperIII.PaperImprovements and PaperIII.PaperImprovementsGate	paper-facing corollaries and selected auxiliary certificates	compiled in the consolidated gate run
PaperIII.PublicAPI, PaperIII.OfPartition, PaperIII.Obstructions	downstream API, split-partition transport, and obstruction certificates	compiled in the recorded closure
FREEZE_REPORT.md and ESCAPE_HATCH_ASSESSMENT.md	immutable build, axiom and raw-token assessment	generated and included in the sealed formal archive

At the SimpleGraph level, Theorem_1_1_of_splitPartition_uncond accepts a finite graph equipped with a clique/independent partition. The construction SplitGraph.ofPartition is connected to the original graph by ofPartitionIso; the vertex count, edge count, ν_3, and Φ are transported through graph-isomorphism invariance, including nu3_congr_of_iso. Other type-specialized theorem forms remain in PaperIII.PublicAPI for downstream use but are not mathematical results of the paper and are not catalogued here.

The directed canonical-module compilation, public- and query-root builds, eight compatible axiom-query files, escape-hatch assessment, immutable source manifests, and internal audit pass their author-side checks. The external audit independently reproduced the public and query roots and verified the designated axiom surfaces against the same frozen bytes.

No computation or regression script is a premise of the proof. The exact-arithmetic and integer-programming checks remain supplementary audits. The completed adversarial review and corpus-bounded prior-art assessment are not substitutes for human peer review or a specialist determination of priority; public tagging is a repository-level release operation.

11.7 Reusable formalization components

The formalization also produced a standalone Contrib.* library whose modules import only Mathlib. Four groups are especially relevant beyond this paper:

- finite packing/covering LP strong duality via a conic-Farkas argument (Contrib.LPStrongDuality);
- variance-sensitive concentration and martingale tails (Contrib.BennettBernstein, Contrib.HoeffdingUpper, Contrib.ConditionalHoeffding, and Contrib.FreedmanBernstein);

- Dirac-type Hamiltonicity and matching results (Contrib.DiracHamiltonian and Contrib.SpreadMatchingDirac);
- finite max-flow/min-cut and selected design-theoretic components (Contrib.MaxFlowMinCut and Contrib.FanoSTS).

These modules are reusable formalization byproducts, not premises of Theorem 1.1. They are held for separate review and possible upstream submission. The present manuscript makes no claim that their APIs or individual statements are novel relative to the current Mathlib library; that comparison requires a separate upstream-oriented literature and API review. Appendix E.6 records only the components closest to the mathematical proof.

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13. Reproducibility

The mathematical source is this manuscript. The manuscript-facing formal surfaces are PaperIII.Theorem_1_1_Final, PaperIII.CanonicalTrianglePacking, PaperIII.PublicAPI, PaperIII.OfPartition, and PaperIII.Obstructions. PaperIII.PaperImprovementsGate is the byproduct gate, and PaperIII.CanonicalTrianglePackingGate checks the exact correspondence between the canonical packing parameters and the Nibble/Yuster model used to discharge AX1.

The formal development is sealed as PAPER_III_lean_v1.4_freeze.zip, with SHA-256 79ee24c38fd776bc2585a0c3c996e30817f0829fc5064463bdbde0fa2d3d7104. It contains 704 Lean source files and the three pinned Lake/toolchain files listed in the 707-entry source manifest. The recorded verification uses Lean 4.28.0 and Mathlib v4.28.0 at commit 8f9d9cff6bd728b17a24e163c9402775d9e6a365.

The available local verification record is summarized below.

Table 3. Sealed local verification record.

Check	Log or command	Current result
public aggregate-root build	gate_logs/run_2026-08-22_CCS_NOTEBOOK456_resumed/logs/02_build_public_root_clean.log	exit 0; 8,455 jobs completed; clean-origin process resumed after an application restart
query-root completion	gate_logs/run_2026-08-22_CCS_NOTEBOOK456_resumed/logs/03_build_query_roots_incremental.log	exit 0; 8,444 jobs completed
canonical statement-and-axiom gate	PaperIII.CanonicalTrianglePackingGate	exit 0; target-side integral and fractional bridges checked
complete theorem-level axiom record	eight FreezeAxioms*.lean files	exit 0; only propext, Classical.choice, and Quot.sound
quantitative Lean-closure regression	FreezeAxiomsAuditClosure.lean and FreezeAxiomsAX1Closure.lean	all named closure surfaces present with foundational-only footprints
escape-hatch assessment	raw-token and focused build/footprint checks	no active sorry warning, no active native_decide, and no project axiom in a canonical surface
immutable seal	source/package manifests, ZIP and SHA-256 sidecar	707 source entries; archive hash verified

The recorded verification used one unchanged source snapshot. Its build sequence was

```
lake build PaperIII

lake build BKLO.MainDenseUnconditional Nibble.AX1Closed PaperIII.CanonicalTrianglePacking PaperIII.Obstruc
tions PaperIII.PaperImprovementsGate PaperIII.PublicAPI PaperIII.Theorem_1_1_Final

#print axioms PaperIII.Theorem_1_1

#print axioms PaperIII.AX1_holds

#print axioms PaperIII.AX2_holds
```

The author-side project began with no .lake directory or compiled project object. The desktop application restarted during the first public-root process; the unchanged project was then resumed incrementally and completed successfully. The immutable freeze therefore records PASS_CLEAN_ORIGIN_RESUMED, not a single uninterrupted author invocation. It preserves that disclosure, verbatim output, component identifiers, source manifest, archive manifest, and SHA-256 inventories. The external audit subsequently supplied an uninterrupted reproduction from an absent project build directory, using the pinned dependency revisions: 8,455 public-root jobs and 8,444 query-root jobs completed with exit code 0, and all eight axiom-query files passed. Those declaration-level checks establish build conformance and axiom footprints; they do not replace independent mathematical rederivation.

Two archived legacy modules, Ax2/PartB/Axioms.lean and Ax2/PartA/Wlog.lean, contain project-local axiom declarations. No source file imports either module, and neither declaration occurs in the dependency footprint of a canonical release surface. Accordingly, the precise formal claim is declaration-level: the canonical theorem and release interfaces have no project-local mathematical axiom; the complete historical source tree is not claimed to contain no axiom token.

The proof itself remains in the manuscript. Regression scripts and computational audits are not hypotheses, reductions, or proof steps.

The canonical repository package contains the complete Lean source, toolchain, manifest, build and axiom logs, manuscript sources, and SHA-256 inventories. The public tag and commit bind those files at repository level; no commit identifier is asserted inside the manuscript itself.

Written materials are intended for distribution under CC BY-NC 4.0. Author-owned verification and regression code is intended for distribution under the PolyForm Noncommercial License 1.0.0; third-party dependencies retain their own licenses.

Review boundary: the independent reproduction and external adversarial audit are complete, and the novelty review is corpus-bounded. Human peer review and specialist priority determination have not been performed. The public tag and commit are repository metadata rather than premises of the theorem.

Appendix C. Computational Audits

C.1 Exact common-profile LP

verify_common_profile_lp.py enumerates the vertices of the symmetrized cover polyhedron using exact rational Gaussian elimination and compares the optimum with (3.5).

C.2 Exact fractional margin

verify_fractional_margin.py checks Theorem 4.2 for

$$3 \le p \le 80, \qquad 1 \le q \le 2p, \qquad 0 \le d \le p$$

using exact rational arithmetic.

C.3 Small ILP audits

verify_factor_rounding.py and verify_shifted_center.py compute exact integral triangle packings on small instances using a binary edge-capacity ILP. They verify the stated upper bounds on $\Phi(G)$.

C.4 Polarization

verify_polarization.py performs exhaustive small-profile and randomized exact-integer checks of Lemma 6.1.

C.5 Divisibility

verify_divisibility_correction.py exhaustively verifies the path parity construction through order eighteen, checks the fixed minimum-degree threshold exactly, and follows the full parity and modulo-three correction on randomized dense residuals.

Remark 2.3′ — Necessity of a density hypothesis

Lean certificates. PaperIII.ax2_divisibility_degree_insufficient, PaperIII.ax2_density_necessary_K7_minus_two_triangles, re-exported through PaperIII.Obstructions.

Corollary 10.2b — Sharp quadratic term and forced linear coefficient

Lean certificate. PaperIII.Corollary_1_2_sharp, assembled from PaperIII.Corollary_1_2, PaperIII.Byproduct_leading_constant_forced, and PaperIII.Byproduct_completeSplit_cp_sharp. The declaration Byproduct_leading_constant_forced certifies the assertion about the linear coefficient; sharpness of the quadratic coefficient follows from the exact complete-split value as $p \to \infty$.

Corollary E.1.1 — Single factor-assignment packing

Lean certificate. Byproduct_factorization_assignment_packing.

Corollary E.1.2 — Double-factorization lower bound

Lean certificate. Byproduct_double_factorization_packing.

Corollary E.1.3 — Reserved-gain packing bound

Lean certificate. Byproduct_reserved_gain_packing_bound_subset.

Corollary E.2.1 — Short-corridor constant 3/2

Lean certificate. Byproduct_short_corridor_constant_three_halves.

Corollary E.2.2 — Packing-to-partition bridge

Lean certificates. Byproduct_cp_le_Phi, Byproduct_low_corridor_cp_three_halves, Byproduct_effective_low_corridor_cp_bound.

Corollary E.3.1 — Exact fractional optimum

Lean certificate. Byproduct_commonProfile_tau3Star_eq_F.

Corollary E.3.2 — Phase regions

Lean certificates. Byproduct_commonProfile_tau3Star_eq_uniform_region, Byproduct_commonProfile_tau3Star_eq_separated_region, Byproduct_commonProfile_tau3Star_eq_hot_region.

Corollary E.4.1 — Fractional slack lower bound

Lean certificate. Byproduct_unified_fractional_margin.

Corollary E.5.1 — Exact value and normal form

Lean certificates. Byproduct_completeSplit_cp_exact_value, Byproduct_completeSplit_cp_sharp.

Corollary E.5.2 — Forced linear coefficient

Lean certificates. Byproduct_leading_constant_forced, packaged with the upper bound and exact family as Corollary_1_2_sharp.

E.6.1 Finite packing/covering LP duality

Lean certificates. Byproduct_triangle_LP_duality; standalone form Contrib.LPDuality.lp_strong_duality in Contrib.LPStrongDuality.

E.6.2 Dirac-type matching and dense correction

Lean certificates. Byproduct_dirac_perfect_matching_even, Byproduct_dirac_near_perfect_matching, Byproduct_dense_divisible_correction_edges; standalone matching form in Contrib.SpreadMatchingDirac.

E.6.3 Uniform nibble interface

Lean certificate. Byproduct_almostPerfectMatching_uniform.

Corollary E.7.1 — Attainment and fractional edge bound

Lean certificates. Byproduct_nu3_achieved, Byproduct_nu3star_le_edges_div_three.

Corollary E.7.2 — Majority-nibble edge-gap form

Lean certificate. Byproduct_nu3star_sub_nu3_le_edge_gap_most.

Appendix E. Selected Formalization-Derived Corollaries

The theorem names follow PaperIII.PaperImprovements and the standalone Contrib.* package. Their release status is governed by the consolidated build and axiom gate in Section 13.

Corollary E.5.2 — Forced linear coefficient

The weaker certificate Byproduct_leading_constant_sharp, which follows immediately from the exact normal form, is not separately catalogued.

E.6.3 Uniform nibble interface

The accompanying standalone library also contains concentration, Hamiltonicity, max-flow/min-cut, codegree-counting, hypergeometric-tail, and design-theoretic modules. Section 11.7 records their scope without presenting them as additional corollaries of this paper.

Corollary 10.2a — Exact complete-split benchmark

The Lean development records the upper bound, lower bound, and equality as Byproduct_completeSplit_cp_le_exact_value, Byproduct_completeSplit_cp_ge_exact_value, and Byproduct_completeSplit_cp_exact_value. Appendix E.5 retains only the exact normal form and the logically distinct forced-linear-coefficient statement.

Packing-form corollaries

The following corollaries restate results from Sections 5 and 7 directly in packing language. They are formalized as factorization_assignment_packing, double_factorization_packing, and reserved_gain_packing_bound_subset. All three are unconditional: none uses Theorem 2.1 or Theorem 2.3.

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